

PRINCE YADAV

Data Scientist | AI/ML Engineer | LLM & RAG Systems

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SUMMARY

Final-year Computer Science student and aspiring AI Engineer specializing in LLM systems, RAG pipelines, and computer vision. Built and shipped end-to-end ML solutions. Comfortable across the full stack: from model training and fine-tuning to Fast API deployment and cloud (Azure). Passionate about building AI that solves real problems, not just benchmarks

TECHNICAL SKILLS

Languages	Python · C++ · SQL
AI / ML	LLMs · Retrieval-Augmented Generation (RAG) · NLP · Computer Vision · Agentic AI · Fine-Tuning (LoRA, QLoRA, RLHF) · Multimodal AI · Time-Series Forecasting · Embeddings · Object Detection · Image Segmentation
Frameworks	PyTorch · TensorFlow · Hugging Face Transformers · LangChain · LangGraph · PEFT · Scikit-learn · OpenCV · Pandas · Streamlit
Databases & cloud	MongoDB · Supabase · Pinecone · ChromaDB · FAISS · Render · GitHub · Azure · Docker & Cloud
Soft Skills	Mentorship & Seminars · Cross-functional Collaboration · Research Communication · Problem Solving

WORK EXPERIENCE

Data Scientist & ML Developer

Jan 2024 – Present

NextAIQLabs · Industrial AI Automation

- Developing an Industrial Automation System using AI and computer vision to streamline and automate real-world manufacturing and inspection workflows.
- Contributing to a child initiative aimed at transforming Indian education — shifting from rote learning to skills-based, practical learning through school-to-school collaboration and hands-on AI exposure.
- Building a unified platform (app + portal) connecting schools, teachers, parents, and admins in one place — enabling end-to-end tracking, management, and collaboration across the entire education system.
- Delivering real-world AI exposure to students through structured programs and direct school partnerships.

AI/ML Intern | ExpensexchangeAI

Jun 2025 – Jul 2025

IBM

- Developed ETL pipelines and time-series forecasting models for a stock portfolio analyzer covering 20+ tickers, achieving 76% directional accuracy and $R^2 = 0.82$.
- Reduced forecast error by 7–10% through Optuna hyperparameter tuning, Huber Loss optimization, and domain-specific feature engineering.
- Practiced MLOps workflows using MLflow for experiment tracking, model versioning, and staging-to-production deployment

PROJECTS

MELD Emotion — Multimodal Emotion Recognition

Jan 2026 – May 2026

End-to-end emotion recognition fusing text, audio, and video modalities via transformer-based fusion.

- Three-modality pipeline: Llama-3.2-3B with LoRA/PEFT (text), CNN-BiLSTM on Mel-spectrograms (audio), 3D ResNet (video) trained on 9,816 samples across 7 emotion classes on HPC GPU cluster.
- Designed TA-AVN cross-modal attention fusion with Squeeze-Excitation gating; addressed severe class imbalance using weighted sampling strategies.

Echo-D — Ultrasound Diagnostic AI

Aug 2025 – Dec 2025

AI-assisted radiology support tool for organ segmentation and automated diagnostic report generation.

- Integrated Meta SAM for organ/lesion segmentation with BLIP-2 for visual question answering, achieving +14% accuracy over U-Net baseline on 500+ anonymized ultrasound scans.
- Containerized the pipeline as a Docker microservice with a modular REST API, enabling zero-downtime model upgrades.

RAG Lab — Advanced RAG System

Jan 2026 – Present

Production-grade retrieval-augmented generation system with hybrid search, reranking, and agentic reasoning for any document type.

- **Architecture:** Engineered hybrid dense + sparse retrieval (ChromaDB + BM25) with Reciprocal Rank Fusion, CrossEncoder reranking, and HyDE query rewriting for precise document question answering.
- Added a LangGraph agentic reasoning loop with tool-calling and self-reflection; validated via RAGAs framework — Faithfulness 0.95, Answer Relevance 0.95. Deployed on Microsoft Azure.
- **Stack:** Fast API async backend, Redis query caching, LangSmith for LLM tracing/eval; React + Tailwind chat UI with source citation and confidence scoring. Supports GPT-4o, Claude 3.5, and local Ollama models.

EDUCATION

B.Tech in Computer Science

Aug 2022 – Present

University of Petroleum and Energy Studies (UPES), Dehradun | CGPA: 7.2 / 10

Coursework: Machine Learning · LLMs & Generative AI · Computer Vision · Explainable AI · Data Structures & Algorithms

CERTIFICATIONS & ACTIVITIES

- 2nd Runner-Up, IBM Hackathon — end-to-end AI solution design under competitive evaluation.
- UPES DSW Ambassador — campus leadership and student-administration

Jan 2024 – feb 2025